



VIMAL JYOTHI
ENGINEERING COLLEGE
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CE-B
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Assignment Cover Page

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Admission No. 7979

Subject Name :- Software Testing Computing

Subject Code:- CIT448

Assignment Title/No : 1

Name of the faculty: Mr. Aravathy Man

Assignment Submitted on 24/02/26

Late submission rules : Max mark will reduced to 50% for 1-5 working day's delay, no mark will be awarded thereafter.

I am hereby confirming that this assignment is my own and I haven't adopted any unfair means in any steps of its preparation to enhance my performance in this assignment.

Date : 24/02/26

Sign with Name

Assignment subdivision	Maximum Mark	Marks awarded	Remarks
A			
B			
C			

Feed back/suggestions :

Name and sign of the faculty

Q. Find the weight required to perform following classification of given input pattern using Hebb rule. the input are "1" where "+" symbol is present and "-1" where "." is present. The "L" pattern belongs to the class (target value +1) and the "U" pattern doesn't belong to the class (target value -1)

$$\begin{array}{ccc}
 + & . & . \\
 + & . & . \\
 + & + & +
 \end{array}
 \quad
 \begin{array}{ccc}
 + & . & + \\
 + & . & + \\
 + & + & +
 \end{array}$$

"L" "U"

Pattern	Inputs										Target y
	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	b	
L	1	-1	-1	1	-1	-1	1	1	1	1	1
U	1	-1	1	1	-1	1	1	1	1	1	-1

Initially $w_1 = w_2 = w_3 = w_4 = w_5 = w_6 = w_7 = w_8 = w_9 = b = 0$

case 1

$$w_i(\text{new}) = w_i(\text{old}) + x_i y$$

$$w_1(\text{new}) = 0 + (1 \times 1) = 1$$

$$w_2(\text{new}) = 0 + (-1 \times 1) = -1$$

$$w_3(\text{new}) = 0 + (-1 \times 1) = -1$$

$$w_4(\text{new}) = 0 + (1 \times 1) = 1$$

$$w_5(\text{new}) = 0 + (-1 \times 1) = -1$$

$$w_6(\text{new}) = 0 + (-1 \times 1) = -1$$

$$w_7(\text{new}) = 0 + (1 \times 1) = 1$$

$$w_8(\text{new}) = 0 + (1 \times 1) = 1$$

$$w_9(\text{new}) = 0 + (1 \times 1) = 1$$

$$b(\text{new}) = b(\text{old}) + y = 0 + 1 = 1$$

$$\Delta w_i = x_i y$$

$$\Delta w_1 = 1 \times 1 = 1$$

$$\Delta w_2 = -1 \times 1 = -1$$

$$\Delta w_3 = -1 \times 1 = -1$$

$$\Delta w_4 = 1 \times 1 = 1$$

$$\Delta w_5 = -1 \times 1 = -1$$

$$\Delta w_6 = -1 \times 1 = -1$$

$$\Delta w_7 = 1 \times 1 = 1$$

$$\Delta w_8 = 1 \times 1 = 1$$

$$\Delta w_9 = 1 \times 1 = 1$$

$$\Delta b = y = 1$$

$$[w_1 \ w_2 \ w_3 \ w_4 \ w_5 \ w_6 \ w_7 \ w_8 \ w_9 \ b] = [1 \ -1 \ -1 \ 1 \ -1 \ -1 \ 1 \ 1 \ 1 \ 0]$$

case II

$$w_1(\text{new}) = 1 + (1 \times -1) = 0$$

$$w_2(\text{new}) = -1 + (-1 \times -1) = 0$$

$$w_3(\text{new}) = -1 + (1 \times -1) = -2$$

$$w_4(\text{new}) = 1 + (1 \times -1) = 0$$

$$w_5(\text{new}) = -1 + (-1 \times -1) = 0$$

$$w_6(\text{new}) = -1 + (1 \times -1) = -2$$

$$w_7(\text{new}) = 1 + (1 \times -1) = 0$$

$$w_8(\text{new}) = 1 + (1 \times -1) = 0$$

$$w_9(\text{new}) = 1 + (1 \times -1) = 0$$

$$\Delta w_1 = 1 \times -1 = -1$$

$$\Delta w_2 = -1 \times -1 = 1$$

$$\Delta w_3 = 1 \times -1 = -1$$

$$\Delta w_4 = 1 \times -1 = -1$$

$$\Delta w_5 = -1 \times -1 = 1$$

$$\Delta w_6 = -1 \times -1 = 1$$

$$\Delta w_7 = 1 \times -1 = -1$$

$$\Delta w_8 = 1 \times -1 = -1$$

$$\Delta w_9 = 1 \times -1 = -1$$

$$\Delta b = y = -1$$

$$[w_1 \ w_2 \ w_3 \ w_4 \ w_5 \ w_6 \ w_7 \ w_8 \ w_9 \ b] = [0 \ 0 \ -2 \ 0 \ 0 \ -2 \ 0 \ 0 \ 0 \ 0]$$

